

Basic Networking Skills for Digital Forensics

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Fellowship in Web Application Security & Digital Forensics

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Lab Scenario

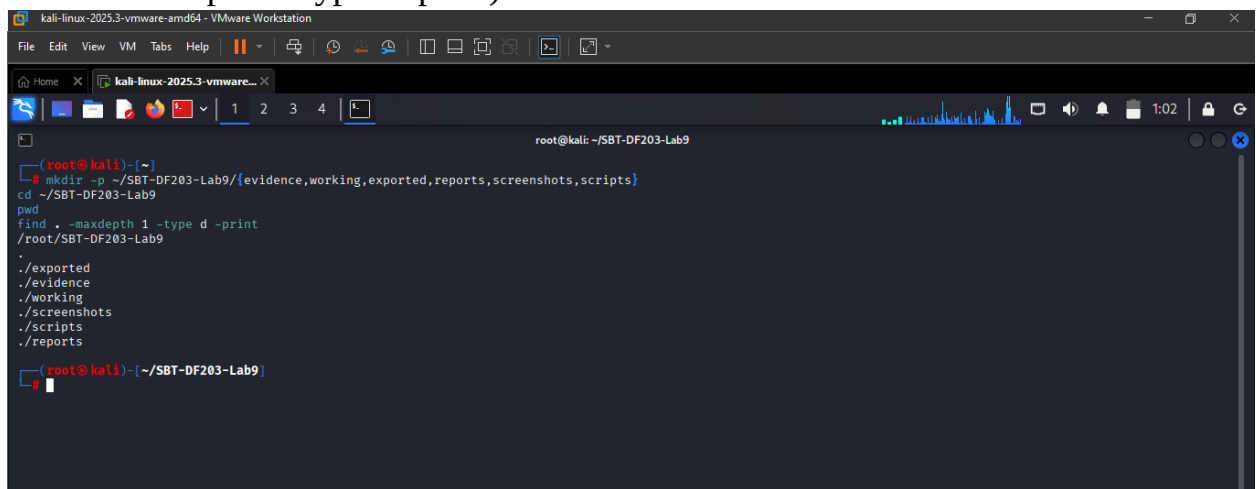
A historical wireless CTF capture protected by deprecated WEP40 has been supplied. You will preserve and decompress the evidence, identify IEEE 802.11 traffic, recover or validate the 40-bit key using Aircrack-ng, decrypt the capture offline, inventory network endpoints, carve transferred files and explain why WEP is cryptographically unsafe.

Objective

The objective of this investigation was to analyze a historical IEEE 802.11 wireless capture protected with WEP40 encryption, identify wireless endpoints, recover or validate the WEP key, decrypt captured traffic, extract transferred objects, and assess the security weaknesses of WEP.

1. Folder Structure (Lab Preparation)

```
(mkdir -p ~/SBT-DF203-  
Lab9/{evidence,working,exported,reports,screenshots,scripts}  
cd ~/SBT-DF203-Lab9  
pwd  
find . -maxdepth 1 -type d -print)
```

A screenshot of a terminal window titled 'kali-linux-2025.3-vmware-amd64 - VMware Workstation'. The terminal shows the following commands and output:

```
root@kali: ~/SBT-DF203-Lab9  
root@kali)~[~]  
# mkdir -p ~/SBT-DF203-Lab9/{evidence,working,exported,reports,screenshots,scripts}  
cd ~/SBT-DF203-Lab9  
pwd  
/root/SBT-DF203-Lab9  
find . -maxdepth 1 -type d -print  
./exported  
./evidence  
./working  
./screenshots  
./scripts  
./reports  
root@kali)~/SBT-DF203-Lab9
```

The terminal window has a dark theme and shows the directory structure being created and listed.

2. Installing packets(sudo apt update

```
sudo apt install -y aircrack-ng wireshark tshark foremost binwalk xz-utils)
```

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
1 2 3 4
root@kali: ~/SBT-DF203-Lab9
(root@kali)~[~/SBT-DF203-Lab9]
# sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [54.1 MB]
Get:4 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:5 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [16.0 kB]
Fetched 75.9 MB in 37s (2.056 kB/s)
2176 packages can be upgraded. Run 'apt list --upgradable' to see them.
wire shark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
The following packages were automatically installed and are no longer required:
  gir1.2-girepository-2.0 libcgmp6 libjs-underscore libmbcrypto16 libraw23t64 openjdk-21-jre python3.13-tk
  imagemagick-7-common libcrypt-dev libjxr-tools libmjpegutils-2.1-0t64 libtsk19t64 openjdk-21-jre-headless python3.13-venv
  libann0 libgavi-1 libjxr0t64 libmpeg2encpp-2.1-0t64 libwireshark18 python3-fs rpcsvc-proto
  libarmadillo14 libgdal37 liblab-gamut1 libmplex2-2.1-0t64 libwiretap15 python3-gpg samba-ad-dc
  libavc1394-0 libgirepository-1.0-1 liblqr-1-0 libpoppler147 libwmflite-0.2-7 python3.13 samba-ad-provision
  libblosc2-4 libpgmep6t64 libmagickcore-7.q16-10 libportmidi0 libwsutil16 python3.13-dev samba-dsdb-modules
  libcdt5 libjs-jquery-ui libmagickwand-7.q16-10 libpython3.13-dev libzixing3 python3.13-minimal
Use 'sudo apt autoremove' to remove them.

Upgrading:
  aircrack-ng binwalk hwloc libhwloc-plugins libhwloc15 liblzma-dev liblzma5 libze1 python3-binwalk xz-utils

Installing:
  foremost

Summary:
  Upgrading: 10, Installing: 1, Removing: 0, Not Upgrading: 2166
  Download size: 1,810 kB / 3,251 kB
  Space needed: 726 kB / 55.7 GB available

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

3. Downloading evidences(wget -O evidence/file.xz \ 'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz')

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
1 2 3 4
root@kali: ~/SBT-DF203-Lab9
(root@kali)~[~/SBT-DF203-Lab9]
# wget -O evidence/file.xz \
'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'
--2026-09-25 01:09:56-- https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)[185.199.108.133]:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12837660 (12M) [application/octet-stream]
Saving to: 'evidence/file.xz'

evidence/file.xz      100%[=====] 12.24M  5.81MB/s  in 2.1s

2026-09-25 01:10:00 (5.81 MB/s) - 'evidence/file.xz' saved [12837660/12837660]

(root@kali)~[~/SBT-DF203-Lab9]
```

4. SHA256sum hashing (sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt)
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6

```
(root@kali)-[~/SBT-DF203-Lab9]
# sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  evidence/file.xz

(root@kali)-[~/SBT-DF203-Lab9]
#
```

5. Creating a working part (cp --preserve=timestamps evidence/file.xz working/file_working.xz)

```
(root@kali)-[~/SBT-DF203-Lab9]
# cp --preserve=timestamps evidence/file.xz working/file_working.xz

(root@kali)-[~/SBT-DF203-Lab9]
#
```

6. Decompressing zip file and verifying file type (unxz -k working/file_working.xz)(file working/file_working)

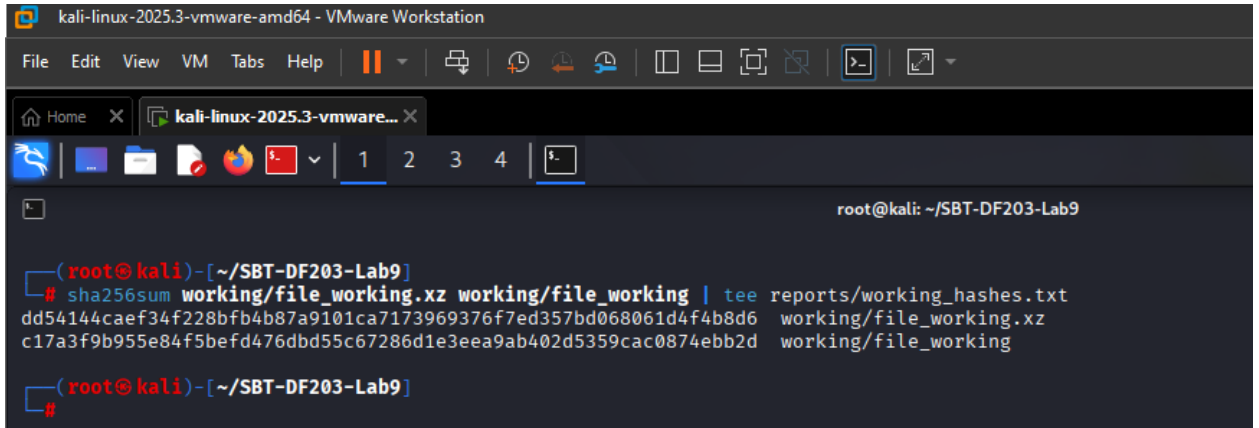
```
(root@kali)-[~/SBT-DF203-Lab9]
# unxz -k working/file_working.xz

(root@kali)-[~/SBT-DF203-Lab9]
#
```

```
(root@kali)-[~/SBT-DF203-Lab9]
# file working/file_working
working/file_working: pcap capture file, microsecond ts (little-endian) - version 2.4 (802.11, capture length 65535)

(root@kali)-[~/SBT-DF203-Lab9]
#
```

7. SHA256 Hashing (sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt).
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6
(working/file_working.xz)
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d
(working/file_working)



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab9
(root@kali)~[~/SBT-DF203-Lab9]
# sha256sum working/file_working.xz | tee reports/working_hashes.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working
(root@kali)~[~/SBT-DF203-Lab9]
#
```

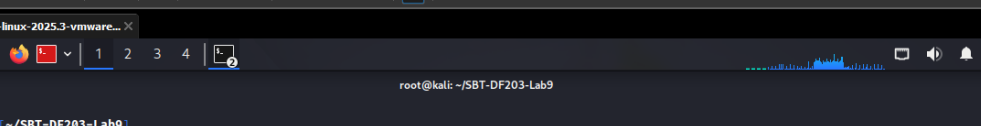
8. Chain of Custody

Field	Entry
Case ID	SBT-DF203-Lab9-Idriss0-Muhamamd-Abdullahi
Examiner	Idriss Muhammad Abdullahi
Date Started	24/09/2026
Evidence File	file.xz
Source	Codegate Historical CTF Dataset
Original SHA256	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6
Working Copy SHA256	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6
Analysis System	Kali Linux VM
Notes	Working copy created before analysis

Part A: Inventory the Wireless Capture

```
(root@kali)~[/SBT-DF203-Lab9]
# capinfos "working/file_working" | tee reports/capinfos.txt
File name:      working/file_working
File type:      Wireshark/tcpdump/... - pcap
File encapsulation: IEEE 802.11 Wireless LAN
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 65535 bytes
Number of packets: 45 k
File size:      14 MB
Data size:      13 MB
Capture duration: 272.969795 seconds
Earliest packet time: 2015-03-06 02:51:54.364525
Latest packet time: 2015-03-06 02:56:27.334320
Data byte rate: 49 kbps
Data bit rate: 395 kbps
Average packet size: 299.13 bytes
Average packet rate: 165 packets/s
SHA256: c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d
SHA1: cb929fd6e407c13639e3168b0f35f921788478d3
Strict time order: False
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = IEEE 802.11 Wireless LAN (20 - ieee-802-11)
    Capture length = 65535
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 45169

(root@kali)~[/SBT-DF203-Lab9]
#
```



The screenshot shows a Kali Linux terminal window with the following content:

```
root@kali: ~/SBT-DF203-Lab9

(root@kali) - [~/SBT-DF203-Lab9]
# tshark -r "working/file_working" -q -z io,phs | tee reports/protocol_hierarchy.txt
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

=====
Protocol Hierarchy Statistics
Filter:

frame                frames:45169 bytes:13511274
  wlan                frames:45169 bytes:13511274
    data              frames:15713 bytes:13104128
=====

(root@kali) - [~/SBT-DF203-Lab9]
```

```
Wlan sample frames (tshark -r " working/file_working" -Y 'wlan' -T fields \
-e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e
wlan.sa -e wlan.da -e wlan.bssid \
| head -n 100 | tee reports/wlan_frame_sample.tsv)
```

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab9

(root@kali)~[~/SBT-DF203-Lab9]
# tshark -r "working/file_working" -Y 'wlan' -T fields \
  -e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e wlan.sa -e wlan.da -e wlan.bssid \
  | head -n 100 | tee reports/wlan_sample.tsv
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
1 2015-03-06T02:51:54.364526000-0500 1 11
2 2015-03-06T02:51:54.364527000-0500 1 11
3 2015-03-06T02:51:54.364525000-0500 1 11
4 2015-03-06T02:51:54.364526000-0500 1 11
5 2015-03-06T02:51:54.365550000-0500 1 11
6 2015-03-06T02:51:54.366063000-0500 1 11
7 2015-03-06T02:51:54.366062000-0500 1 11
8 2015-03-06T02:51:54.366061000-0500 1 11
9 2015-03-06T02:51:54.366063000-0500 1 11
10 2015-03-06T02:51:54.366061000-0500 1 11
11 2015-03-06T02:51:54.366064000-0500 1 11
12 2015-03-06T02:51:54.366063000-0500 1 11
13 2015-03-06T02:51:54.366063000-0500 1 11
14 2015-03-06T02:51:54.366063000-0500 1 11
15 2015-03-06T02:51:54.366064000-0500 1 11
16 2015-03-06T02:51:54.366062000-0500 1 11
17 2015-03-06T02:51:54.366575000-0500 1 11
18 2015-03-06T02:51:54.366575000-0500 1 11
19 2015-03-06T02:51:54.366573000-0500 1 11
20 2015-03-06T02:51:54.366574000-0500 1 11
21 2015-03-06T02:51:54.366575000-0500 1 11
22 2015-03-06T02:51:54.366575000-0500 1 11
23 2015-03-06T02:51:54.366575000-0500 1 11
24 2015-03-06T02:51:54.366574000-0500 1 11
25 2015-03-06T02:51:54.366574000-0500 1 11
26 2015-03-06T02:51:54.366574000-0500 1 11
39 2015-03-06T02:51:54.366575000-0500 1 11
40 2015-03-06T02:51:54.366574000-0500 1 11
41 2015-03-06T02:51:54.366573000-0500 1 11
42 2015-03-06T02:51:54.366573000-0500 1 11
43 2015-03-06T02:51:54.366574000-0500 1 11
44 2015-03-06T02:51:54.367085000-0500 1 11
45 2015-03-06T02:51:54.367086000-0500 1 11
46 2015-03-06T02:51:54.367087000-0500 1 11
47 2015-03-06T02:51:54.367089000-0500 1 13
48 2015-03-06T02:51:54.367085000-0500 1 11
49 2015-03-06T02:51:54.367087000-0500 1 11
50 2015-03-06T02:51:54.367087000-0500 1 11
51 2015-03-06T02:51:54.367087000-0500 1 11
52 2015-03-06T02:51:54.367087000-0500 1 11
53 2015-03-06T02:51:54.367086000-0500 1 11
54 2015-03-06T02:51:54.367633000-0500 0 8 00:26:66:55:97:d6 ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6
55 2015-03-06T02:51:54.368111000-0500 1 11
56 2015-03-06T02:51:54.368111000-0500 1 11
57 2015-03-06T02:51:54.368110000-0500 1 11
58 2015-03-06T02:51:54.368110000-0500 1 11
59 2015-03-06T02:51:54.368110000-0500 1 11
60 2015-03-06T02:51:54.368110000-0500 1 11
61 2015-03-06T02:51:54.368622000-0500 1 11
62 2015-03-06T02:51:54.368623000-0500 1 11
63 2015-03-06T02:51:54.368621000-0500 1 11
64 2015-03-06T02:51:54.368622000-0500 1 11
65 2015-03-06T02:51:54.368621000-0500 1 11
66 2015-03-06T02:51:54.368623000-0500 1 11
67 2015-03-06T02:51:54.368622000-0500 1 11
68 2015-03-06T02:51:54.368623000-0500 1 11
69 2015-03-06T02:51:54.368622000-0500 1 11
70 2015-03-06T02:51:54.368622000-0500 1 11
71 2015-03-06T02:51:54.368623000-0500 1 11

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

Part B: Identifying WEP Protection and Key Parameters

Locating protected frames(tshark -r "working/file_working" -Y 'wlan.fc.protected==1' -T fields \ -e frame.number -e frame.time_epoch -e wlan.sa -e wlan.da -e wlan.bssid -e wlan.wep.iv -e wlan.wep.key \ | head -n 200 | tee reports/wep_protected_frames.tsv)

```

(root@kali)~[~/SBT-DF203-Lab9]
# tshark -i working/file_working -Y 'wlan.fc.protected==1' -T fields \
  -e frame.number -e frame.time_epoch -e wlan.sa -e wlan.da -e wlan.bssid -e wlan.wep.iv -e wlan.wep.key \
  | head -n 200 | tee reports/wep_protected_frames.tsv
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
9432 1425628319.232982000 00:26:66:55:97:d4 ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x67ffec 0
10130 1425628320.138252000 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 00:26:66:55:97:d6 0xc2cbeb 0
10155 1425628320.176656000 f0:f6:1c:68:96:7c 04:8d:38:48:e5:b4 00:26:66:55:97:d6 0xc3cbeb 0
10157 1425628320.177168000 f0:f6:1c:68:96:7c 04:8d:38:48:a8:b5 00:26:66:55:97:d6 0xc4cbeb 0
10159 1425628320.177168000 f0:f6:1c:68:96:7c 08:10:77:92:7c:2f 00:26:66:55:97:d6 0xc5cbeb 0
10161 1425628320.177168000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0xc6cbeb 0
10163 1425628320.177168000 f0:f6:1c:68:96:7c c8:3a:35:58:c4:ba 00:26:66:55:97:d6 0xc7cbeb 0
10165 1425628320.177167000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0xc8cbeb 0
10167 1425628320.177686000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0x68ffec 0
10170 1425628320.180760000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x69ffec 0
10171 1425628320.182288000 f0:f6:1c:68:96:7c 33:33:00:00:00:02 00:26:66:55:97:d6 0xc9cbeb 0
10172 1425628320.187417000 00:26:66:55:97:d4 ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x6affec 0
10173 1425628320.187408000 f0:f6:1c:68:96:7c 33:33:00:00:00:02 00:26:66:55:97:d6 0xc9cbeb 0
10175 1425628320.187919000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0xcacbeb 0
10177 1425628320.188952000 f0:f6:1c:68:96:7c 33:33:00:00:00:02 00:26:66:55:97:d6 0x6bffec 0
10178 1425628320.189976000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x6cffec 0
10183 1425628320.192016000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0xcbcbeb 0
10185 1425628320.193048000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x6dffec 0
10224 1425628320.259088000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0xcccbeb 0
10228 1425628320.262680000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x6effec 0
10229 1425628320.262678000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0x6ffffec 0
10233 1425628320.263184000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0xcdcbeb 0
10261 1425628320.419350000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x6ccbeb 0
10266 1425628320.420374000 f0:f6:1c:68:96:7c 01:00:5e:00:00:02 00:26:66:55:97:d6 0xcfcbeb 0
10268 1425628320.420374000 f0:f6:1c:68:96:7c 01:00:5e:00:00:fb 00:26:66:55:97:d6 0xd0cbeb 0
10270 1425628320.420886000 f0:f6:1c:68:96:7c 01:00:5e:00:00:02 00:26:66:55:97:d6 0x70ffec 0

```

WEP traffic with source MAC addresses such as f0:f6:1c:68:96:7c and BSSID 00:26:66:55:97:d6. The IV field contains values such as 0xc2cbeb, 0xc5cbeb, 0xc9cbeb, and 0x6ffec. The displayed WEP key/index field is 0. 64-bit WEP uses a 24-bit Initialization Vector (IV) combined with a 40-bit secret key, giving a nominal 64-bit total key size.

IVs and RC4 weaknesses

WEP is insecure because its relatively small 24-bit IV can repeat, especially on busy networks. Repeated IVs allow attackers to collect enough encrypted traffic to exploit weaknesses in the RC4-based WEP design and potentially recover the secret key. Therefore, WEP should not be used for modern wireless security.

```

tshark -r "working/file_working" -Y 'wlan.fc.protected==1' -T fields -e
wlan.wep.iv \
  | sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt

```

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab9

(root@kali)~/~/SBT-DF203-Lab9
# tshark -i "working/file_working" -Y "wlan.fc.protected==1" -e wlan.wep.iv \
# sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
7 0x42e8eb
7 0x28d8eb
6 0x95dfeb
5 0x9415ed
5 0x86e5eb
5 0x7ae5eb
5 0x28e7eb
5 0x19e5eb
4 0xd5d7eb
4 0xd3e8eb
4 0xc915ed
4 0xc7d8eb
4 0xad4eb
4 0x73e5eb
4 0x65d8eb
4 0x4ad5eb
4 0x31e7eb
4 0x2723ed
4 0x10e7eb
3 0xffe5eb
3 0xf609ed
3 0xa307ed
3 0x97dbeb
3 0x96d1eb
3 0x92e8eb
3 0x91e8eb
```

Part C - Recover or Validate the WEP40 Key

aircrack-ng "working/file_working" | tee reports/aircrack_output.txt

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab9

(root@kali)~/~/SBT-DF203-Lab9
# aircrack-ng "working/file_working" | tee reports/aircrack_output.txt
Reading packets, please wait ...
Opening working/file_working
Read 45169 packets.
Got 15477 out of 15000 IVsStarting PTW attack with 15477 ivs.

# BSSID          ESSID          Encryption
1 00:26:66:55:97:D6 cgnetwork      WEP (15477 IVs)

Choosing first network as target.

Reading packets, please wait ...
Opening working/file_working
Read 45169 packets.

1 potential targets

Attack will be restarted every 5000 captured ivs.
```

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...
root@kali: ~/SBT-DF203-Lab9
Aircrack-ng 1.7
[00:00:00] Tested 83 keys (got 15477 IVs)
KB depth bytes(vote)
0 0/ 1 A4(22784) 62(20992) A0(19968) B6(19968) 42(19456) 6E(19456) 01(19200) B7(19200) 26(18944) 60(18944) 9E(18944) 0E(18688) 74(18688)
1 0/ 1 3D(23840) 51(20736) 07(20480) 62(19968) 78(19968) 1F(19712) 80(19712) 8D(19456) 80(19200) 85(19200) 9D(19200) 13(18944) EC(18944)
2 0/ 1 F6(23808) E4(20992) D0(20736) 68(20224) 95(19712) 38(19456) 0C(19200) 45(18944) 4F(18944) A8(18944) F7(18944) 60(18688) 63(18688)
3 1/ 10 F3(20480) C5(19968) D0(19968) 3E(19712) 43(19456) 52(19456) B2(19456) 09(19456) 20(19456) 8F(19200) A0(19200) B0(19200) 04(18944)
4 6/ 9 01(19712) 20(19456) 3E(19456) 52(19456) 5C(19456) 90(19456) F9(19456) 45(18944) 85(18944) 95(18944) D3(18944) 3A(18688) 40(18688)
KEY FOUND! [ A4:3D:F6:F3:74 ]
Decrypted correctly: 100%
(root@kali)-[~/SBT-DF203-Lab9]
```

`printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt`

```
(root@kali)-[~/SBT-DF203-Lab9]
# printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt
A4:3D:F6:F3:74
(root@kali)-[~/SBT-DF203-Lab9]
#
```

Part D - Decrypt the Capture Offline

`airdecap-ng -w A43DF6F374 file_working | tee
../reports/airdecap_output.txt`

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab9/working

(root@kali)-[~/SBT-DF203-Lab9]
# cd working

(root@kali)-[~/SBT-DF203-Lab9/working]
# airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
Total number of stations seen      10
Total number of packets read      45169
Total number of WEP data packets  15477
Total number of WPA data packets   0
Number of plaintext data packets   0
Number of decrypted WEP packets    15477
Number of corrupted WEP packets     0
Number of decrypted WPA packets     0
Number of bad TKIP (WPA) packets    0
Number of bad CCMP (WPA) packets    0

(root@kali)-[~/SBT-DF203-Lab9/working]
#
```

find working -maxdepth 1 -type f -ls | tee
reports/decrypted_files_inventory.txt

```
(root@kali)-[~/SBT-DF203-Lab9]
# find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
524947  13904  -rw-rw-r--  1 root    root      14234002 Sep 25 01:10 working/file_working
524958  12488  -rw-rw-r--  1 root    root      12784501 Sep 25 03:49 working/file_working-dec
524946  12540  -rw-rw-r--  1 root    root      12837660 Sep 25 01:10 working/file_working.xz

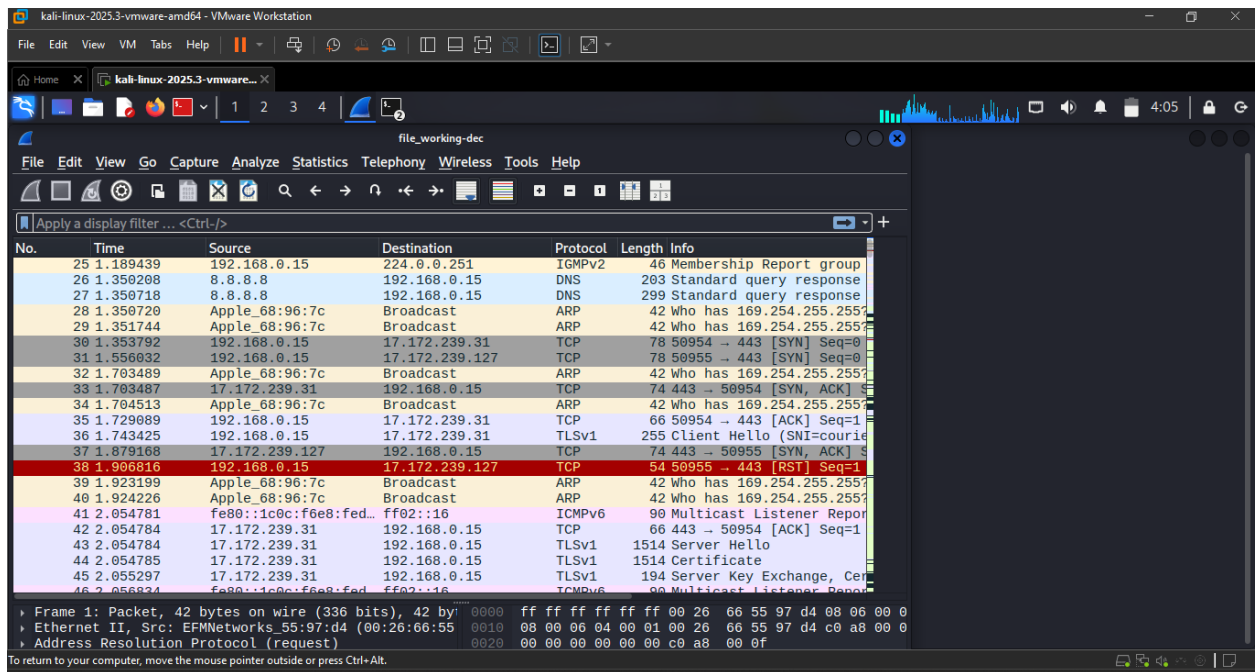
(root@kali)-[~/SBT-DF203-Lab9]
#
```

sha256sum working/* | tee reports/all_working_file_hashes.txt

```
(root@kali)-[~/SBT-DF203-Lab9]
# sha256sum working/* | tee reports/all_working_file_hashes.txt
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d  working/file_working
167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04  working/file_working-dec
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  working/file_working.xz

(root@kali)-[~/SBT-DF203-Lab9]
#
```

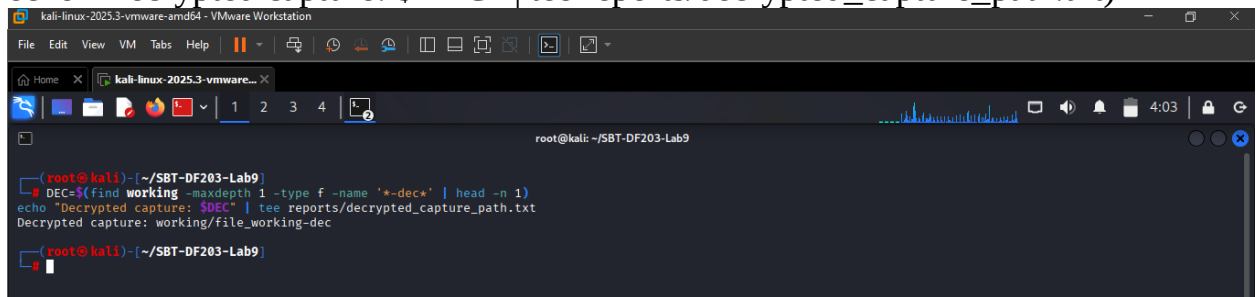
IP/TCP/HTTP or other higher-layer protocols are now visible.



Part E - Extract Endpoints and Conversations

(DEC=\$(find working -maxdepth 1 -type f -name '*-dec*' | head -n 1)

echo "Decrypted capture: \$DEC" | tee reports/decrypted_capture_path.txt)



tshark -r "\$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...
root@kali: ~/SBT-DF203-Lab9

(root@kali)~[~/SBT-DF203-Lab9]
# tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

Ethernet Endpoints
Filter:<No Filter>

| Packets | | Bytes | | Tx Packets | | Tx Bytes | | Rx Packets | | Rx Bytes |
Apple_68:96:7c 15407 12 MB 7076 616 kB 8331 11 MB
EFMNetworks_55:97:d4 15020 12 MB 8178 11 MB 6842 593 kB
ASUSTekCOMPU_2a:c2:7a 358 281 kB 223 270 kB 135 11 kB
IPv4mcast_fb 42 6,648 bytes 0 0 bytes 42 6,648 bytes
IPv6mcast_fb 36 7,022 bytes 0 0 bytes 36 7,022 bytes
Broadcast 31 3,386 bytes 0 0 bytes 31 3,386 bytes
IPv4mcast_7f:ff:fa 31 11 kB 0 0 bytes 31 11 kB
IPv6mcast_16 10 920 bytes 0 0 bytes 10 920 bytes
IPv6mcast_02 9 630 bytes 0 0 bytes 9 630 bytes
IPv4mcast_16 3 180 bytes 0 0 bytes 3 180 bytes
IPv4mcast_02 2 92 bytes 0 0 bytes 2 92 bytes
NetisTechnol_48:e5:b4 1 42 bytes 0 0 bytes 1 42 bytes
NetisTechnol_48:a8:b5 1 42 bytes 0 0 bytes 1 42 bytes
08:10:77:92:7c:2f 1 42 bytes 0 0 bytes 1 42 bytes
TendaTechnol_58:c4:ba 1 42 bytes 0 0 bytes 1 42 bytes
IPv6mcast_ff:2a:c2:7a 1 78 bytes 0 0 bytes 1 78 bytes
```

tshark -r "\$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...
root@kali: ~/SBT-DF203-Lab9

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

IPv4 Endpoints
Filter:<No Filter>

| Packets | | Bytes | | Tx Packets | | Tx Bytes | | Rx Packets | | Rx Bytes |
192.168.0.15 15309 12 MB 6995 608 kB 8314 11 MB
198.90.20.111 8057 6,916 kB 4446 6,637 kB 3611 279 kB
199.27.79.193 5694 4,825 kB 3131 4,603 kB 2563 221 kB
192.168.0.9 329 277 kB 197 265 kB 132 11 kB
173.194.127.141 153 109 kB 80 102 kB 73 7,364 bytes
173.194.127.55 143 89 kB 71 82 kB 72 6,990 bytes
8.8.8.8 81 10 kB 40 6,960 bytes 41 3,391 bytes
199.16.158.186 66 15 kB 29 10 kB 37 5,214 bytes
173.194.127.227 66 24 kB 26 15 kB 40 8,455 bytes
173.194.127.229 55 22 kB 23 13 kB 32 9,731 bytes
31.13.70.1 51 19 kB 26 15 kB 25 3,151 bytes
173.241.248.220 44 6,848 bytes 20 3,122 bytes 24 3,726 bytes
198.41.208.140 44 28 kB 21 25 kB 23 2,718 bytes
224.0.0.251 42 6,648 bytes 0 0 bytes 42 6,648 bytes
173.194.127.237 36 9,526 bytes 14 6,435 bytes 22 3,091 bytes
216.58.220.234 34 15 kB 16 12 kB 18 2,497 bytes
192.168.0.1 33 12 kB 33 12 kB 0 0 bytes
173.241.248.219 33 5,733 bytes 14 2,807 bytes 19 2,926 bytes
104.74.158.12 33 21 kB 18 20 kB 15 1,236 bytes
74.125.203.156 32 20 kB 16 17 kB 16 2,534 bytes
239.255.255.250 31 11 kB 0 0 bytes 31 11 kB
173.194.127.231 31 6,816 bytes 12 5,197 bytes 19 1,619 bytes
198.41.209.137 28 15 kB 13 14 kB 15 1,556 bytes
17.172.239.31 25 6,504 bytes 12 3,973 bytes 13 2,531 bytes
173.194.72.95 23 4,077 bytes 9 2,232 bytes 14 1,845 bytes
173.194.127.122 21 12 kB 11 11 kB 10 1,759 bytes
69.59.169.185 18 5,644 bytes 7 3,675 bytes 11 1,969 bytes
```

tshark -r "\$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab9

(root@kali)~[~/SBT-DF203-Lab9]
# tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

TCP Conversations
Filter:<No Filter>

| Frames | Bytes | | Frames | Bytes | | Total | Relative | Duration |
|         |         | |         |         | |         | Start     |          |
192.168.0.15:51007 ↔ 198.90.20.111:80 | 1257 | 1,885 kB | 978 | 73 kB | 2235 | 1,958 kB | 113.588792000 | 45.4035
192.168.0.15:50959 ↔ 199.27.79.193:80 | 1209 | 1,816 kB | 959 | 71 kB | 2168 | 1,887 kB | 21.074236000 | 34.8633
192.168.0.15:51008 ↔ 198.90.20.111:80 | 1159 | 1,739 kB | 947 | 70 kB | 2106 | 1,810 kB | 113.588792000 | 46.0179
192.168.0.15:51001 ↔ 199.27.79.193:80 | 1145 | 1,725 kB | 927 | 66 kB | 2072 | 1,791 kB | 63.066042000 | 26.6088
192.168.0.15:51009 ↔ 198.90.20.111:80 | 687 | 1,028 kB | 572 | 44 kB | 1259 | 1,072 kB | 113.605175000 | 28.7983
192.168.0.15:51014 ↔ 198.90.20.111:80 | 664 | 991 kB | 566 | 44 kB | 1230 | 1,036 kB | 114.020474000 | 25.0538
192.168.0.15:51006 ↔ 198.90.20.111:80 | 654 | 970 kB | 520 | 40 kB | 1174 | 1,011 kB | 113.578039000 | 25.4942
192.168.0.15:51000 ↔ 199.27.79.193:80 | 375 | 561 kB | 323 | 25 kB | 698 | 586 kB | 58.731642000 | 4.4552
192.168.0.15:50999 ↔ 199.27.79.193:80 | 206 | 302 kB | 180 | 15 kB | 386 | 319 kB | 56.792636000 | 1.9390
192.168.0.15:51036 ↔ 192.168.0.9:5000 | 98 | 144 kB | 70 | 5,705 bytes | 168 | 150 kB | 243.325633000 | 0.1777
192.168.0.15:50958 ↔ 199.27.79.193:80 | 73 | 93 kB | 67 | 12 kB | 140 | 105 kB | 20.145401000 | 14.9846
192.168.0.15:51035 ↔ 192.168.0.9:5000 | 80 | 116 kB | 56 | 4,756 bytes | 136 | 121 kB | 243.281602000 | 0.1398
192.168.0.15:50983 ↔ 173.194.127.141:80 | 71 | 99 kB | 59 | 4,392 bytes | 130 | 104 kB | 27.831547000 | 31.4188
192.168.0.15:50997 ↔ 199.27.79.193:80 | 62 | 87 kB | 37 | 3,826 bytes | 99 | 90 kB | 56.126972000 | 11.0492
192.168.0.15:51034 ↔ 173.194.127.229:443 | 23 | 13 kB | 32 | 9,731 bytes | 55 | 22 kB | 234.384001000 | 6.7568
192.168.0.15:51005 ↔ 198.90.20.111:80 | 25 | 22 kB | 28 | 4,757 bytes | 53 | 27 kB | 113.019956000 | 2.8111
192.168.0.15:50960 ↔ 31.13.70.1:443 | 26 | 15 kB | 25 | 3,151 bytes | 51 | 19 kB | 21.329724000 | 32.8418
192.168.0.15:51028 ↔ 173.194.127.227:443 | 18 | 11 kB | 29 | 7,603 bytes | 47 | 18 kB | 172.217100000 | 37.9452
192.168.0.15:51003 ↔ 198.41.208.140:80 | 21 | 25 kB | 23 | 2,718 bytes | 44 | 28 kB | 102.780857000 | 32.8878
192.168.0.15:51018 ↔ 173.194.127.55:80 | 20 | 24 kB | 16 | 1,494 bytes | 36 | 26 kB | 115.102391000 | 30.4286
192.168.0.15:50973 ↔ 199.16.158.186:443 | 15 | 5,156 bytes | 19 | 2,640 bytes | 34 | 7,796 bytes | 25.665885000 | 31.5233
192.168.0.15:51002 ↔ 104.74.158.12:80 | 18 | 20 kB | 15 | 1,236 bytes | 33 | 21 kB | 63.175899000 | 30.4746
192.168.0.15:50974 ↔ 199.16.158.186:443 | 14 | 5,082 bytes | 18 | 2,574 bytes | 32 | 7,656 bytes | 25.665885000 | 31.5223

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```

tshark -r "\$DEC" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.col.Protocol \

| head -n 200 | tee reports/ip_mac_mapping_sample.tsv

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab9

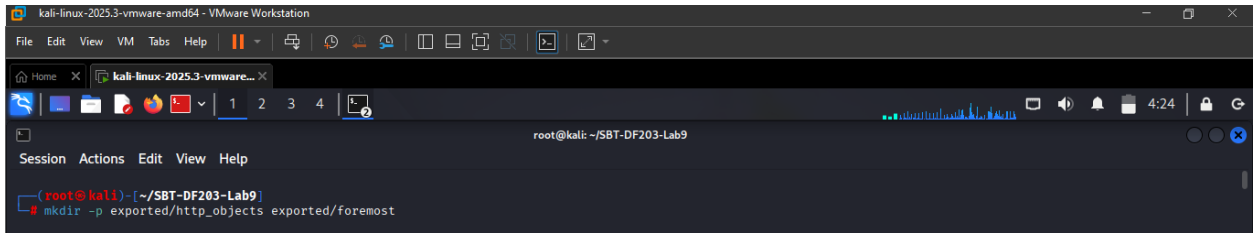
(root@kali)~[~/SBT-DF203-Lab9]
# tshark -r "$DEC" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.col.Protocol \
| head -n 200 | tee reports/ip_mac_mapping_sample.tsv
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
7 0.0.0.0 255.255.255.255 DHCP
9 0.0.0.0 255.255.255.255 DHCP
11 192.168.0.1 255.255.255.255 DHCP
20 192.168.0.15 8.8.8.8 DNS
21 192.168.0.15 8.8.8.8 DNS
22 192.168.0.15 224.0.0.2 IGMPv2
23 192.168.0.15 224.0.0.251 IGMPv2
24 192.168.0.15 224.0.0.2 IGMPv2
25 192.168.0.15 224.0.0.251 IGMPv2
26 8.8.8.8 192.168.0.15 DNS
27 8.8.8.8 192.168.0.15 DNS
30 192.168.0.15 17.172.239.31 TCP
31 192.168.0.15 17.172.239.127 TCP
33 17.172.239.31 192.168.0.15 TCP
35 192.168.0.15 17.172.239.31 TCP
36 192.168.0.15 17.172.239.31 TLSv1
37 17.172.239.127 192.168.0.15 TCP
38 192.168.0.15 17.172.239.127 TCP
42 17.172.239.31 192.168.0.15 TCP
43 17.172.239.31 192.168.0.15 TLSv1
44 17.172.239.31 192.168.0.15 TLSv1
45 17.172.239.31 192.168.0.15 TLSv1
47 0.0.0.0 255.255.255.255 DHCP
48 0.0.0.0 255.255.255.255 DHCP
49 192.168.0.15 17.172.239.31 TCP
50 192.168.0.15 17.172.239.31 TCP
51 192.168.0.1 192.168.0.15 DHCP

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

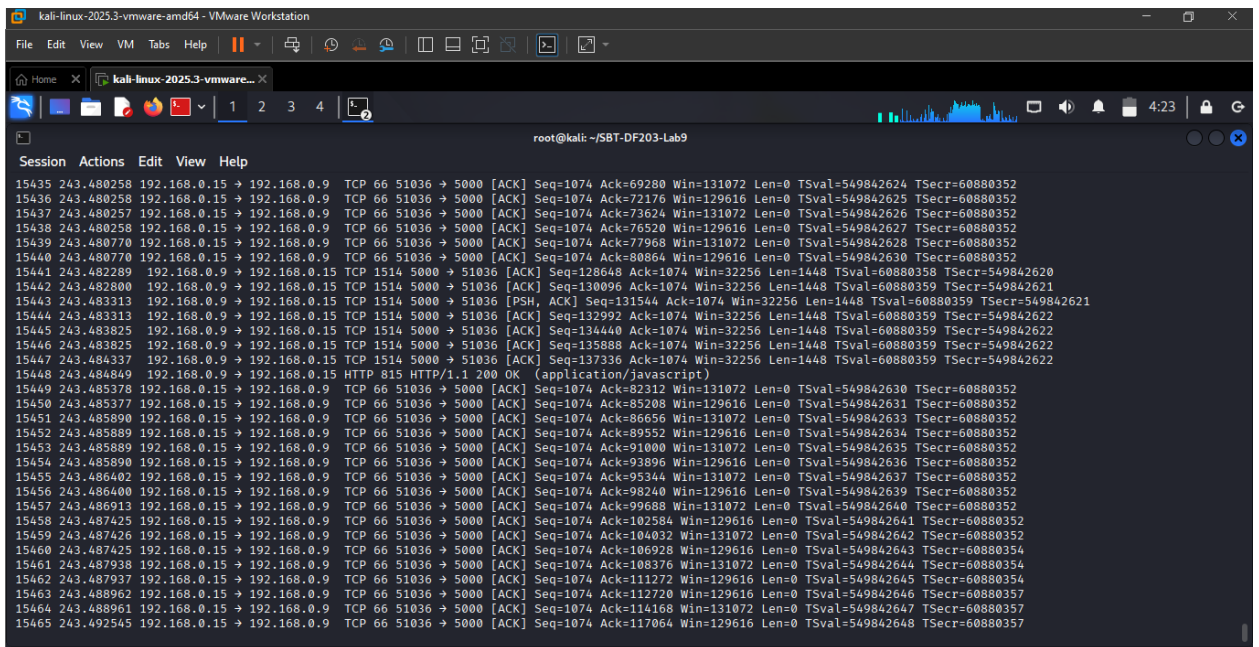
```

Part F - Extract Images, HTML and Other Objects

Created an export folder (mkdir -p exported/http_objects exported/foremost)



(tshark -r "\$DEC" \ --export-objects http,exported/http_objects \ 2>&1 | tee reports/http_export_log.txt)



Generic Carving

foremost -i "\$DEC" -o exported/foremost \ | tee reports/foremost_log.txt
find exported -type f -exec file {} \; | tee reports/exported_file_types.txt

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
1 2 3 4
root@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help
(root@kali)~[~/SBT-DF203-Lab9]
# foremost -i "SOEC" -o exported/foremost \
| tee reports/foremost_log.txt
Processing: working/file_working-dec
[*]

(root@kali)~[~/SBT-DF203-Lab9]
# find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
exported/http_objects/ogli5PA.gifv: HTML document, ASCII text
exported/http_objects/hls%3fversion=4.12: XML 1.0 document, ASCII text
exported/http_objects/v2(2): JSON text data
exported/http_objects/dynamik-min.css: ASCII text, with very long lines (55362)
exported/http_objects/gauid=5372002708res=320x568x326p6ch=UTF-86tz=-5406ws=300x250: ASCII text
exported/http_objects/bootstrap-responsive.min.css: ASCII text, with very long lines (16608)
exported/http_objects/header-contact.png: PNG image data, 19 x 19, 8-bit/color RGBA, non-interlaced
exported/http_objects/ads%3fgdfr_req=18correlator=39741610776330246output=json_html6callback=callbackProxy8impl=fifs6json_a=16eid=1088090356sfv=1-0-26iu_parts=4761660%2CH_I_300x250enc_prev_ius=%2F0%2F16prev_iu_szs=300x25.14055857036ga_sid=14256284806ga_hid=218395456ga_fc=true: JavaScript source, Unicode text, UTF-8 text, with very long lines (34622), with no line terminators
exported/http_objects/object6689.application%2fjavascript: HTML document, ASCII text, with CRLF line terminators
exported/http_objects/of_destiny.png%3fv=h%2Bw3qHw8NrIcs96j8ZN%2BqruehW91KTg3Y9jHy9f3jgs243%2FhnENQoVopWm1m2Z2LH6T7UGA4V5EX3D: PNG image data, 1 x 1, 8-bit/color RGBA, non-interlaced
exported/http_objects/YDAoLskQQ5MOAgvHUQcCLQa6gm6b500u2Qn-iPLo1Go.woff: Web Open Font Format, TrueType, length 22824, version 1.1
exported/http_objects/ogli5PA(1).mp4: data
exported/http_objects/cs.js: empty
exported/http_objects/figure-26.png: PNG image data, 634 x 209, 8-bit/color RGBA, non-interlaced
exported/http_objects/object6681.application%2fjavascript: JavaScript source, ASCII text, with very long lines (3768)
exported/http_objects/lumbar.gif%3fa=x78k%2UID%22%22ae1543dcdbab603f50e1849b3db2c79e9ce5872c0878dfc88d838afac1a3d%22%22accountId%22%23Anull%22%22sessionId%22%23AK24d0d205eeaf1f11bd5b09802b761dc%22%22platform%22%23AK22web%22%22source%22%23AK(3).com%2F8iVy7t6%22%2D: ASCII text, with no line terminators
exported/http_objects/rombo.png: PNG image data, 61 x 70, 8-bit/color RGBA, non-interlaced
exported/http_objects/hit(1).json: JSON text data
```

find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
1 2 3 4
root@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help
(root@kali)~[~/SBT-DF203-Lab9]
# find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt
a3cbb4cd0761b71263bb964e5e2b99367c6e90d3e64a2ae6aa78d96bf603faf0 exported/http_objects/ogli5PA.gifv
0868db67f79acd281fcb3b46c9768dae7033be91f9551185394abf19d4b797ae exported/http_objects/hls%3fversion=4.12
a840281fc37d697b1515a380e500c690be86927a8a6dbb2bcc071447c6b27031 exported/http_objects/v2(2)
fe4529ba03075f4ba177e0898b91def18bad106bb55a1ef4a2560bf982ceaff8 exported/http_objects/dynamik-min.css
187f2ad17d1fee508625e4be5b0d35c7f6cde4209d458a3717f739a25860e4 exported/http_objects/gauid=5372002708res=320x568x326p6ch=UTF-86tz=-5406ws=300x250
9a8149b07d046c4a606e185ad68acdad7695d532a781f556ac86f081c1e5f341 exported/http_objects/bootstrap-responsive.min.css
792477b4d204cbf0580011ca406dd92425e4e42ec5d22c11b4b41e06ac90676 exported/http_objects/header-contact.png
d4726b0714d6ca5c346bf985b11fb8841cd6135372b084843c2b6a863db30 exported/http_objects/ads%3fgdfr_req=18correlator=39741610776330246output=json_html6callback=callback
Proxy8impl=fifs6json_a=16eid=1088090356sfv=1-0-26iu_parts=4761660%2CH_I_300x250enc_prev_ius=%2F0%2F16prev_iu_szs=300x25.14055857036ga_sid=14256284806ga_hid=218395456ga_fc=true
9936530c477845cdda0828673cab8f3ad0cf9cf589123a6e9d15413e1fd1a083 exported/http_objects/object6689.application%2fjavascript
ebf4f635a17d10d6eb646a680b70142419aa3220f228001a036d311a22ee9d2a exported/http_objects/of_destiny.png%3fv=h%2Bw3qHw8NrIcs96j8ZN%2BqruehW91KTg3Y9jHy9f3jgs243%2FhnENQoV
opWm1m2Z2LH6T7UGA4V5EX3D
e97ffa450c3e5b483f0084536b170f97d9c5c6cf096c8223a6b332c861376b exported/http_objects/YDAoLskQQ5MOAgvHUQcCLQa6gm6b500u2Qn-iPLo1Go.woff
96a296d224f285c67be93c30f8a309157f0daa35dc5b87e410b78630a09cfc7 exported/http_objects/ogli5PA(1).mp4
e3b0c44298f1c194abf4c8996fb92427ae41e4649b934ca495991b7852b855 exported/http_objects/cs.js
18ab31de23688c95cc5402b86156512b3c4b747ecd5ad91937b30bf6e6284b exported/http_objects/figure-26.png
64ac226d737e521fc20e69e663b040d195c677050ae66d72a3d83a5c148ee84 exported/http_objects/object6681.application%2fjavascript
74234e98afe7498f1b5d4f1f36ac2d78ac2c339464f950703b0c019892f982b90b exported/http_objects/lumbar.gif%3fa=x78k%2UID%22%22ae1543dcdbab603f50e1849b3db2c79e9ce5872c0878d
ffc8d838afac1a3d%22%22accountId%22%23Anull%22%22sessionId%22%23AK24d0d205eeaf1f11bd5b09802b761dc%22%22platform%22%23AK22web%22%22source%22%23AK(3).com%2F8iVy7t6%22%2D
716%22%2D
eb5c89dca6a23a1f03550d1c749ef0027b48a644bc371e3a560e005495928c73 exported/http_objects/rombo.png
902a974466764157ac302761ad4bf7241655853ec46f8b4e3d0db87023b900b4 exported/http_objects/hit(1).json
1e81216e67645c4363da811ba786e4b6a68aa850e43765323ef58b5293e6d78 exported/http_objects/object6488
a330a0196da3072001cf3494d5e3ae7bd7ed26db7214a24e9f7488d2e7e9d54 exported/http_objects/nivo-slider.css
20550f7bcb2a817ac9a5879e04260da8268e971c0b8031a6b7a2f48a55ee60d5 exported/http_objects/superfish.args.min.js
0ba081f546084b5097aa8a73c75931d5aa1fc4d6e846e53c21f986e6a1509988 exported/http_objects/jquery.1.10.2.min.js
ee1c79d0b533cc5ff0ac1f99d84fbcd7dc7e07acbe9defcac65416b6387950d exported/http_objects/mobile_unified_button_icon_white.png
```

WEP Security Analysis

1. WEP uses RC4 encryption with a short 24-bit IV.
2. Repeated IVs cause keystream reuse.
3. Statistical attacks enable key recovery.
4. Integrity protection is weak and vulnerable to replay attacks.

Conclusion

The historical WEP-protected capture was successfully examined and decrypted. Wireless endpoints were identified, encrypted traffic was recovered, and transferred objects were extracted. The investigation demonstrates why WEP should no longer be used in modern wireless environments.